

Introduction to a Small New Keynesian Model for Policy Analysis

Ministry of Finance and Economic Planning, Rwanda
May 2021

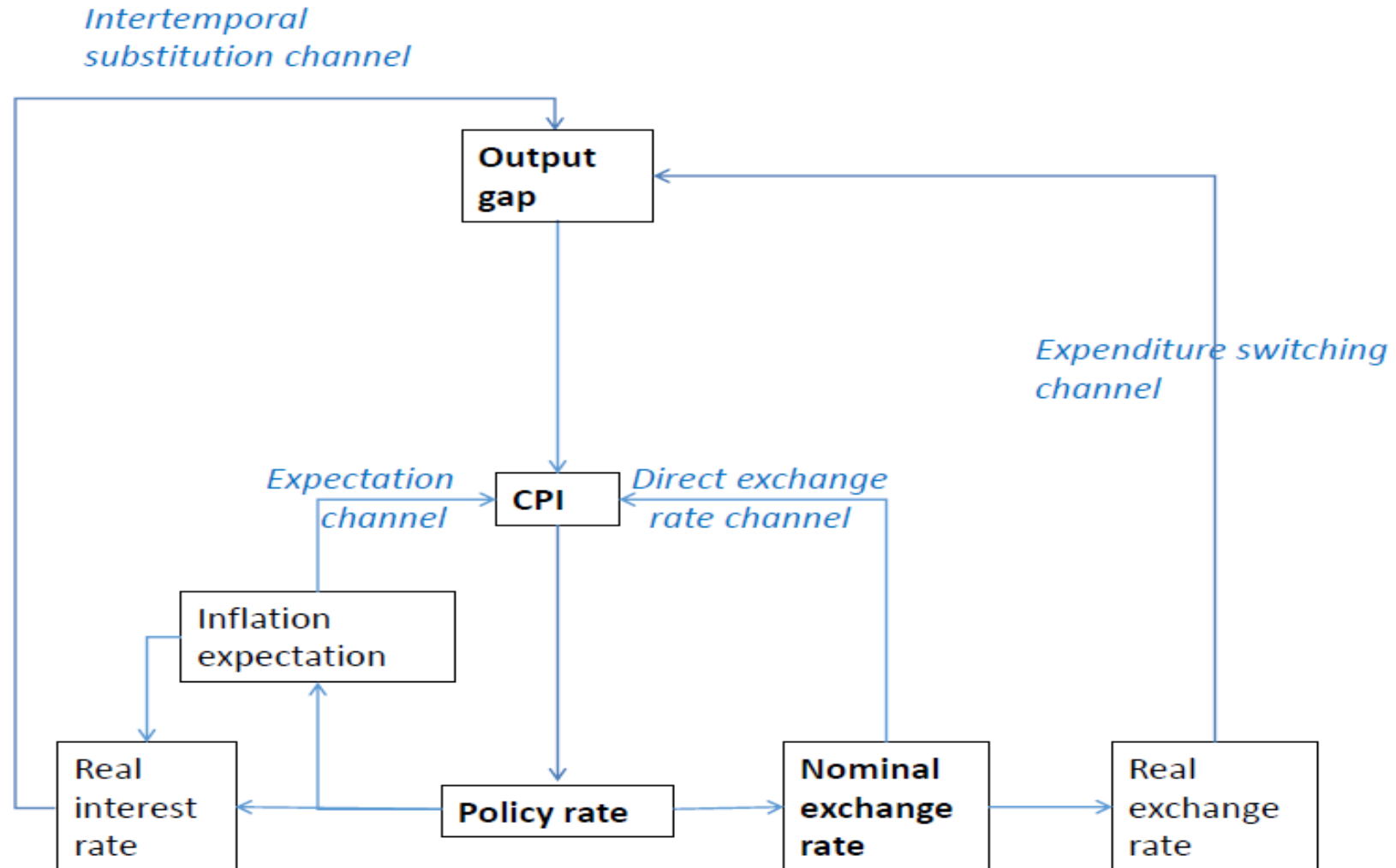
New Keynesian model origin

- The model is sometimes called a “New Keynesian reduced-form model”
 - Ideas of monopolistic competition and sticky-prices/wages
 - Equations resemble log-linearized equations of a micro-founded model
- But:
 - Some parts are ad-hoc (do not have micro-foundations) in order to fit data
 - Equation coefficients are not derived from deep parameters of the micro-founded model, but are estimated/calibrated

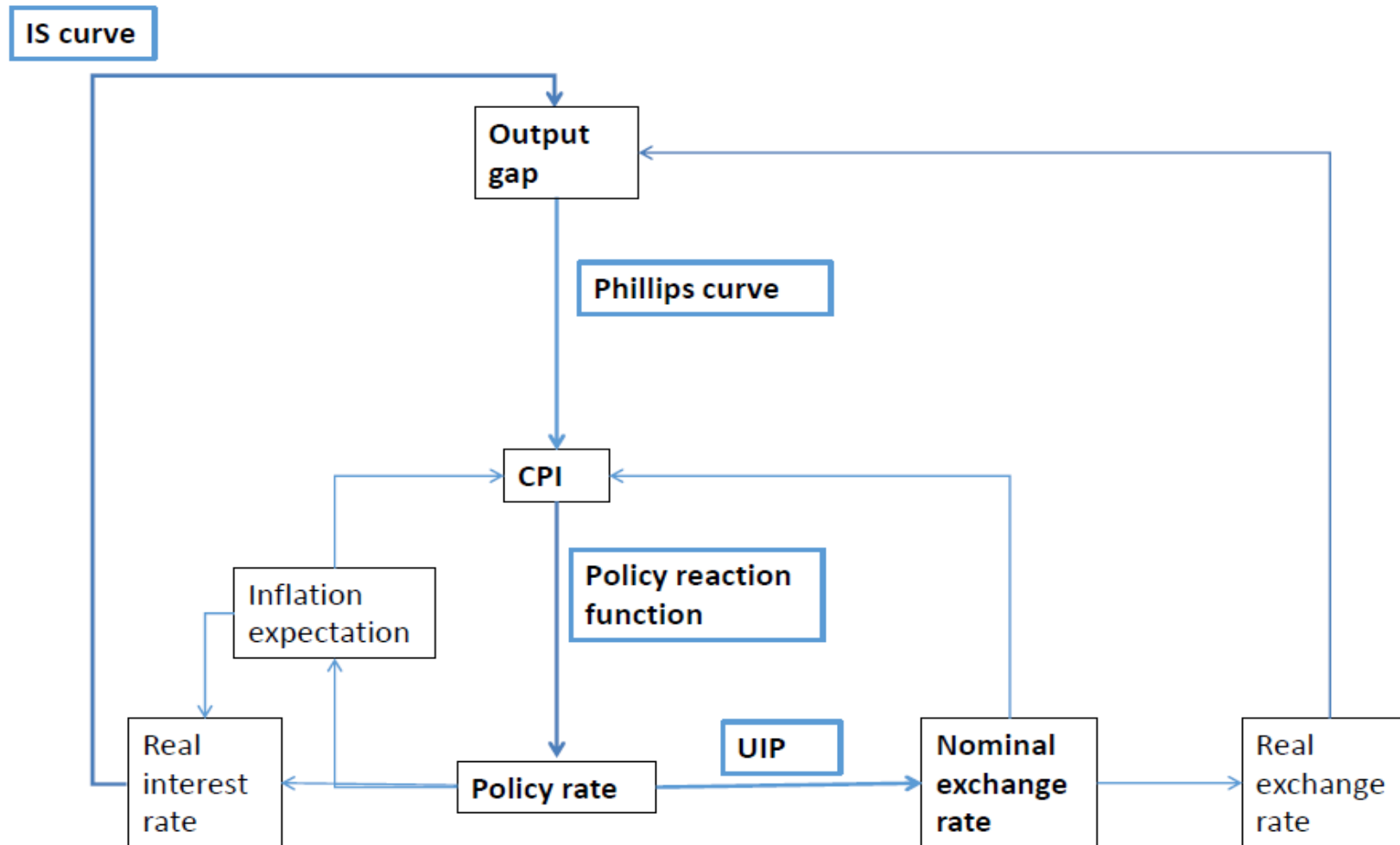
A canonical Model: Building Blocks

- Key behavioral equations
 - Aggregate demand
 - Depends negatively on the real interest rate and positively on real exchange rate
 - Aggregate supply
 - Excess demand leads to higher inflation.
 - Policy rule
 - Sets the interest rate in order to stabilize the nominal path of the economy.
- Basic features for an open economy
 - Uncovered interest rate parity
 - Exchange rate is determined by interest rate differentials.
 - Country-specific extensions

A canonical Model: Key mechanisms and variables



A canonical Model: Key mechanisms and variables



MODEL EQUATIONS

Aggregate demand

- Aggregate demand equation is inspired by the reduced form of the Euler equation with ad-hoc elements, e.g. foreign demand
- Derived from consumer's optimization problem
- Utility maximization:
$$\max_{c_t, l_t} E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, l_t)$$
 - Note: infinite horizons with declining importance (beta)
- Where
$$U(c_t, l_t) = \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \frac{l_t^{1+\varphi}}{1+\varphi} \right]$$
 - Period utility depends on consumption (+) and labor (-)

Aggregate demand - continued

- Constraint: $c_t + Inv_t + S_t + T_t = w_t l_t + r_t^k k_t$

$$Inv_t = k_{t+1} - (1 - \sigma) k_t$$

$$S_t = B_t - (1 + r_t) B_{t-1}$$

- First order condition (Euler equation):

$$U'(c_t) = \beta E_t[(1 + r_t)U'(c_{t+1})]$$

$$c_t^{-\sigma} = \beta E_t[(1 + r_t)c_{t+1}^{-\sigma}]$$

- Intuitively, the marginal utility of consumption today equals the expected marginal utility of consumption tomorrow.
- Defines the dynamic path (growth rate) of consumption, depending on the incentives (r) and on preferences (β , σ)
 - Higher $r \rightarrow$ more saving / less consumption today

Aggregate demand – still continued

Adding habit: utility doesn't depend on contemporaneous consumption only, but on habits (past patterns) as well. Let the utility function be:

$$U(c_t, l_t) = \left[\frac{H_t^{1-\sigma}}{1-\sigma} - \frac{l_t^{1+\varphi}}{1+\varphi} \right]$$

$$H_t = c_t - \gamma c_{t-1}$$

where γ measures the force of habit

Aggregate demand - continued

- The log-linearized Euler equation is derived from the households' optimization (with habit formation type of utility)

$$\hat{c}_t = \left(\frac{1}{1+\gamma} \right) E_t \hat{c}_{t+1} + \left(\frac{\gamma}{1+\gamma} \right) \hat{c}_{t-1} - \frac{1}{\sigma(1+\gamma)} (\hat{i}_t - E_t \pi_{t+1})$$

- Sum across all agents' consumption and express in terms of output gap $\hat{y}_t$ to get the IS curve

$$\hat{y}_t = \left(\frac{1}{1+\gamma} \right) E_t \hat{y}_{t+1} + \left(\frac{\gamma}{1+\gamma} \right) \hat{y}_{t-1} - \frac{1}{\sigma(1+\gamma)} \hat{r}_t \quad \hat{r}_t = \hat{i}_t - E\{\pi_{t+1}\} - r_t^n$$

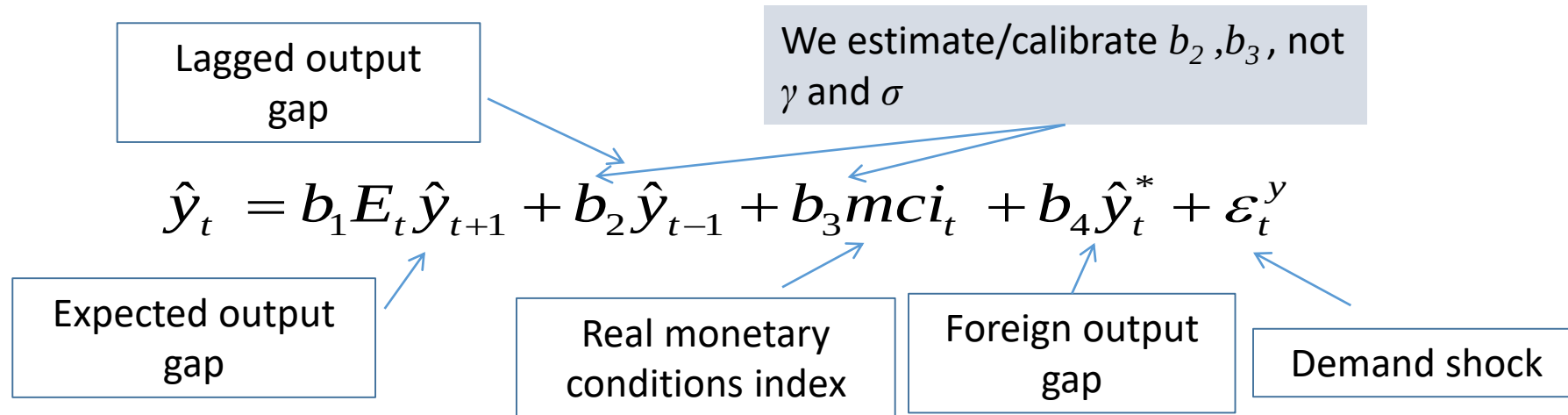
where r_t^n is a neutral (equilibrium) level of the interest rate

γ (controls the strength of habit formation) and σ (drives risk aversion) are deep structural parameters

- Demand equation in our model is similar ...

Aggregate demand – finally there

- Relates monetary policy and real economic activity (output gap):



- foreign output is an ad-hoc element - $\hat{y}_t^*$
- mci is a monetary conditions index: for now it only includes i -rate gap, later on we will also add e -rate gap (ad-hoc):

$$mci_t = \hat{r}_t$$

Aggregate supply, inflation

- Relates real economic activity and inflation
- Derivation from producer's price setting problem
 - Perfect competition: price = marginal cost
 - We assume monopolistic competition: price = $(1 + \text{mark-up}) * \text{marginal cost}$
- Problem: finding the optimal price if firms are not allowed to change prices every period (implicit menu costs)
 - Flexible prices: constant mark-up
 - Rigid prices: variable mark-ups
- Intuition to the solution: all future marginal costs will matter

Aggregate supply - continued

Price setting problem:

- Objective of the producer – maximum profit: $\max_{P_t(j)} E_0 \sum_{t=0}^{\infty} \Pi_t(j)$
- Subject to natural production constraints: $\Pi_t(j) = P_t(j) \cdot y_t(j) - C(q_t(j))$
- And demand curve: $y_t(j) = \left(\frac{P_t}{P_t(j)} \right)^{\theta} y_t$

Aggregate supply - continued

Solution to the price setting problem:

- Optimal price is (1 + mark-up) over marginal cost:
$$P_t(j) = \frac{\theta}{\theta-1} MC(y_t(j))$$
$$P_t(j) = (1 + \mu) MC(y_t(j))$$
- Marginal cost is a bit tricky:
$$MC(y_t(j)) = \frac{1}{A_t \alpha^\alpha (1-\alpha)^{(1-\alpha)}} (R_t^k)^\alpha (W_t)^{1-\alpha}$$
- Log-linearized:
$$\widetilde{mc}_t = \alpha \widetilde{r}_t^k + (1 - \alpha) \widetilde{w}_t - \widetilde{A}_t$$

Aggregate supply - continued

Introducing price stickyness:

- Changing prices is costly, so not optimal to react to all shocks
- Extension of Calvo pricing: Christiano, Eichenbaum and Evans (2005)
 - Each period, firms adjust prices:
 - With probability $(1 - \omega)$ firm chooses a new price level price
 - With probability ω firm indexes based on past inflation

$$P_t = \omega P_{t-1} + (1 - \omega)P_t^*$$

- Degree of price stickiness is given by ω
- Flexible price case is $\omega = 0$

Aggregate supply - continued

Price setting: intuition

- Price setting gets forward-looking if prices are sticky
- All future expected marginal costs count in the optimal price (with decreasing weights going forward)
- Firms forecast future business cycles and set prices so that they keep their desired markups constant *on average*
- Increase in AD in the future should be compensated by setting a higher price today
- **Expectations play a crucial role in price setting!**

Aggregate supply - continued

Optimal price setting under sticky prices :

- Phillips curve:

$$\tilde{\pi}_t = \beta E_t \tilde{\pi}_{t+1} + \frac{(1-\gamma)(1-\gamma\beta)}{\gamma} \tilde{mc}_t + \varepsilon_t^{\tilde{\pi}} \approx \beta E_t \tilde{\pi}_{t+1} + \frac{(1-\gamma)(1-\gamma\beta)}{\gamma} \hat{y}_t + \varepsilon_t^{\tilde{\pi}}$$

- Expected future BC fluctuations: reflected fully in current inflation
- Unexpected future BC fluctuations: reflected partly in future inflation

Aggregate supply - continued

From Phillips curve to CPI equation:

- Adding persistence: backward-looking firms base their pricing expectations on the previous period price adjustments
 - Inflation is persistent
 - Disinflation is costly (more inertia = bigger sacrifice ratio)

We calibrate b_1 and b_2 directly,
not β and ω (deep structural parameters)

Cost push shock

Later on we will add
exchange rate gap
here as well

$$\pi_t = a_1 \pi_{t-1} + (1 - a_1) E\{\pi_{t+1}\} + a_2 rmc_t + \varepsilon_t^\pi$$

$$rmc_t = \hat{y}_t$$

Lagged value

Expected value

Real marginal costs

Policy rule

- The policy rule is added to close the NK model
- It should ensure unique equilibrium, in particular, in the rule...

$$\hat{i}_t = \delta \pi_t + v_t$$

- ... the “Taylor principle” suggests $\delta > 1$
 - the nominal interest rates should increase more than one-to-one in response to inflation
- The most common empirical “Taylor rule”:

$$\hat{i}_t = \delta_{\pi} \pi_t + \delta_x \hat{y}_t + v_t$$

- An extension is “Forward looking Taylor rule”:

$$\hat{i}_t = \delta_{\pi} E_t \{ \pi_{t+1} \} + \delta_x x_t + v_t$$

Model: Interest Rate Policy Rule

- IR rule is the central bank reaction function

$$i_t = g_1 i_{t-1} + (1 - g_1)(i_t^n + g_2 (E_t \{\pi_{t+4}\} - \pi_{t+4}^T)) + \varepsilon_t^i$$

Diagram illustrating the components of the Interest Rate Policy Rule equation:

- Smoothing component (g_1)
- Neutral interest rate (i_t^n)
- Deviation of expected inflation from the target ($E_t \{\pi_{t+4}\} - \pi_{t+4}^T$)
- Monetary Policy Shock (ε_t^i)

- Neutral nominal rate = trend real interest rate + model-consistent inflation expectations

$$i_t^n = \bar{r} + E_t \{\pi_{t+4}\}$$

- The lag is added to account for the degree of “wait-and-see” when setting the policy rate

Full 'helloworld' model: we begin with only the most essential equations

- Aggregate demand

$$\hat{y}_t = \alpha_1 * \hat{y}_{t-1} - \alpha_3 * \hat{r}_t + \varepsilon_t^y$$

- Aggregate supply

$$\pi_t = \beta_1 \pi_{t-1} + (1 - \beta_1) \pi_{t+1}^e + \beta_2 y_t + \varepsilon_t^\pi$$

- MP reaction function

$$i_t = \gamma_1 i_{t-1} + (1 - \gamma_1) [(\bar{r} + \bar{\pi}) + \gamma_2 (\pi_{t+4} - \bar{\pi})] + \varepsilon_t^i$$

- Real interest rate

$$r_t = i_t - \pi_{t+1}$$

- Real rate gap

$$\hat{r}_t = r_t - \bar{r}$$

Full 'helloworld' model: also adding a fiscal block

- Expanded IS equation $\hat{y}_t = \alpha_1 * \hat{y}_{t-1} - \alpha_3 * \hat{r}_t + \alpha_4 * \text{fisc_imp}_t + \varepsilon_t^y$
- Overall fiscal impulse $\text{fisc_imp}_t = d3 * \text{def_discr}_t + (1 - d3) * (\text{def_str}_t - \text{def_str}_{t-1})$
- Discretionary fiscal impulse $\text{def_discr}_t = d2 * \text{def_discr}_{t-1} + \varepsilon_t^{\text{def_discr}}$
- Structural fiscal impulse $\text{def_str}_t = d4 * \text{def_str}_{t-1} + \varepsilon_t^{\text{def_str}}$